Foundations of Arithmetic Power Geometry V:

Discrete Moduli-Stack Formulations, Scale-Dependent Metrics, and Boundary Obstruction Limits

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Abstract

We present the finalized structural architecture of Arithmetic Power Geometry (APG) over discrete parameter spaces. Moving past ill-defined real-variable trajectories and continuous conductor functions, this work establishes an ordered arithmetic chain of surfaces over the universal moduli stack of elliptic curves $\mathcal{M}_{\text{ell}}$. We evaluate the Absolute Projective Closure Defect $\mathcal{D}_p$ alongside a scale-dependent Logarithmic Scale-Weighted Functional $\mathcal{D}_p^*(a,b)$ designed to track the analytic height floor against the classic geometric Szpiro ceiling. We establish the pointwise positivity of the discrete closure defect over primitive integer supports, removing invalid uniform lower bounds. To address the isotropic baseline, we implement a localized algebraic parity obstruction exploring how specific local ramification profiles at the prime $q = 2$ interact with modular level-lowering constraints. Finally, we execute an exact symbol-by-symbol asymptotic boundary analysis under extreme coordinate skewness ($w_a \to 0$). We prove that the scale-weighted height floor stabilizes safely at a true, non-contradictory identity ($0 \leq 4$), confirming that continuous APG deformations, when discretized correctly, reach a non-contradictory containment boundary rather than reproducing the Ribet–Wiles fracture. This locates the definitive boundary limits of continuous geometric methods in exponential Diophantine analysis.

Keywords: arithmetic geometry; discrete parameter spaces; Shannon entropy; Faltings height; Frey curve; Szpiro inequality; Galois representations; modularity lifting.

1 Introduction and Architectural Corrections

The interplay between discrete Galois representations and continuous analytic manifolds remains a central bottleneck in modern arithmetic geometry (Akhtar, 2026c; Darmon et al., 1997; Hecke, 1928). While the foundational proof of the modularity conjecture for semi-stable elliptic curves relied strictly on rigid, discrete ring deformations (Taylor & Wiles, 1995; Wiles, 1995), alternative exploratory frameworks have sought to model exponent variations as parameter deformations across abstract manifolds (Akhtar, 2026a, 2026b). Early iterations of Arithmetic Power Geometry (APG) introduced continuous trajectories $t \in [2, p]$ to track algebraic closure defects relative to a classical Euclidean baseline (Akhtar, 2026a, 2026c).

However, these continuous formulations introduced severe mathematical anomalies:

1. **The Radical Deficit:** Evaluating a number-theoretic conductor function

$$N(t) = \text{rad}(a^t b^t c^t)$$

over continuous real parameters $t \notin \mathbb{Z}$ is algebraic fiction, as prime factorization does not exist over $\mathbb{R}$ (Borevich & Shafarevich, 1966; Lang, 2002).

2. **The Metric Collapse:** Under extreme coordinate skews ($w_a \to 0$), the scale-invariant Shannon entropy functional approaches zero ($H(W) \to 0$), causing the integrated geometric floor to sink beneath the arithmetic ceiling (Akhtar, 2026d; Cover & Thomas, 2006).

This manuscript presents the regularized, discrete incarnation of APG, designated as Volume V. We systematically eliminate real-variable paths, introduce a scale-weighted functional to isolate coordinate dimensions, and map out the definitive boundary lines where continuous analytic structures stabilize against the rigid parameters of the Ribet–Wiles regime (Akhtar, 2026d; Ribet, 1990; Serre, 1987).

2 Discrete Parameter Landscapes and Structural Weight Metrics

Let $(a, b, c) \in (\mathbb{Z}^+)^3$ be a primitive, pairwise coprime integer solution tuple satisfying a hypothetical Fermat relation

$$a^p + b^p = c^p$$

at an odd prime exponent $p > 2$ (Frey, 1986; Hellegouarch, 1972). To maintain strict arithmetic definition, we banish real-parameter trajectories entirely (Akhtar, 2026c).

Definition 1. *The discrete parameter space is an ordered arithmetic chain of primes indexed exclusively by integer sequences (Akhtar, 2026c):*

$$\mathcal{S}_{\mathbb{P}} = \{3, 5, 7, 11, \ldots, p\}.$$

Definition 2. *For any level slice $p_k \in \mathcal{S}_{\mathbb{P}}$, the scale-invariant distribution of the coordinate components is mapped onto the unit interval via the normalized coordinate weights established in Akhtar (2026a):*

$$w_a = \frac{a^2}{a^2 + b^2}, \qquad w_b = \frac{b^2}{a^2 + b^2},$$

where $w_a + w_b = 1$. The information content of this specific configuration is tracked via its corresponding coordinate entropy (Cover & Thomas, 2006; Shannon, 1948):

$$H(W) = -(w_a \ln w_a + w_b \ln w_b).$$

Theorem 1 (Pointwise Positivity of the Discrete Closure Defect)**.** *Let*

$$\mathcal{D}_p(a, b) := 1 - \left(w_a^{p/2} + w_b^{p/2}\right)$$

be the absolute projective closure defect functional evaluated at the discrete target prime p. For any fixed, non-trivial primitive integer support over $\mathbb{Z}^+$, the functional satisfies

$$\mathcal{D}_p(a, b) > 0$$

pointwise (Akhtar, 2026c).

Proof. We evaluate the boundary constraints of the coordinate weight space.

1. **The Isotropic Baseline** ($w_a = w_b = \frac{1}{2}$):

$$\mathcal{D}_p\left(\frac{1}{2}, \frac{1}{2}\right) = 1 - 2\left(\frac{1}{2}\right)^{p/2} = 1 - 2^{1-p/2}.$$

For all odd primes $p \geq 3$,

$$1 - 2^{1-p/2} \geq 1 - \frac{1}{\sqrt{2}} \approx 0.2929 > 0.$$

2. **The Asymptotic Skew Boundary** ($w_a \to 0,\ w_b \to 1$):

$$\lim_{w_a \to 0} \mathcal{D}_p(a, b) = 1 - \left(0 + 1^{p/2}\right) = 0.$$

Because a, b, c are fixed integers within a given triple, w_a cannot reach the continuous limit of 0 for non-trivial configurations (Bombieri & Gubler, 2006). For any fixed, concrete coordinate set, the minimum non-zero value is bounded by

$$w_a = \frac{1}{1 + b^2}.$$

Crucially, because b can grow arbitrarily large across the infinite set of all potential integer pairs, this relation does not establish a universal, uniform lower bound away from zero. It guarantees strict pointwise positivity $\mathcal{D}_p(a, b) > 0$ for every individual primitive pair, preventing local functional collapse while allowing asymptotic decay under infinite coordinate skews (Bugeaud, 2004). $\square$

3 Modular Curve Geometry and Regularized Operator Kernels

To interface with classical modular curves without metric tearing, each prime step $p_k \in \mathcal{S}_{\mathbb{P}}$ is mapped directly to a discrete level slice (Akhtar, 2026b; Shimura, 1971).

Definition 3. *For each step in the arithmetic chain, the discrete modular conductor N is evaluated as the integer radical factor of the coprime products (Deligne, 1971; Lang, 1983):*

$$N = \mathrm{rad}(abc) \in \mathbb{Z}^+.$$

Because $\mathrm{rad}(x^{p_k}) = \mathrm{rad}(x)$ for all positive integers and prime exponents, the conductor remains completely invariant of the exponent level slice, guaranteeing that the corresponding congruence subgroup $\Gamma_0(N) \subset SL_2(\mathbb{Z})$ is rigidly fixed across the entire arithmetic chain:

$$\Gamma_0(N) = \left\{ \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \in SL_2(\mathbb{Z}) \,\middle|\, \gamma \equiv 0 \pmod{N} \right\}.$$

Theorem 2 (Resolvent Well-Definedness Away from the Spectrum)**.** *The spectral resolvent operator kernel*

$$R_s(z, z') = [\Delta_k - s(s-1)]^{-1}$$

attached to the discrete hyperbolic modular curve

$$X_0(N) = \mathbb{H}/\Gamma_0(N)$$

is rigorously well-defined for all parameters s chosen outside the active spectrum.

Proof. Because the conductor N evaluates strictly to a positive integer, the corresponding algebraic curve can be compactified via Deligne–Mumford methods into a proper scheme $\mathcal{X}_0(N)^*$ (Abbes & Ullmo, 1997; Hartshorne, 1977). Prior to compactification, noncompact quotients feature a continuous spectrum which gives rise to specialized analytic behaviors at the boundaries.

By mapping the system onto the chosen regularized compact model, the relevant operator is considered away from spectral poles. General spectral theory guarantees that the spectrum of the Laplace–Beltrami operator on a compactified arithmetic surface is discrete and non-negative (Saito, 1988). Therefore, the resolvent operator does not suffer from singular boundary collisions and remains completely well-defined for all parameters s chosen outside this spectrum (Gross & Zagier, 1986). ☐

4 Upgraded Scale-Dependent Height Bottlenecks

To track the scale parameters of the continuous paths against arithmetic dimensions, we introduce an absolute logarithmic scale factor directly into the analytic functional core.

Definition 4. *The Logarithmic Scale-Weighted Defect functional $\mathcal{D}_p^*(a, b)$ is defined as (Akhtar, 2026d):*

$$\mathcal{D}_p^*(a, b) := \mathcal{D}_p(a, b) \cdot \ln(abc).$$

The existence of a rigorous APG–Arakelov height coupling theorem remains an open problem; consequently, the following bottleneck relation is stated conditionally.

Proposition 1 (The Conditional Scale-Insulated Bottleneck)**.** *Assuming the APG–Arakelov matching coefficient is normalized as stated, the total metric self-intersection of the relative canonical sheaf over the discrete arithmetic chain establishes the conditional bottleneck lower boundary (Faltings, 1984; Szpiro, 1981):*

$$h_F(E) \geq \frac{p \cdot \mathcal{D}_p^*(a, b) \cdot H(W)}{24},$$

which interfaces with the standard conditional Szpiro-type ceiling to generate the arithmetic bottleneck:

$$\frac{p \cdot \mathcal{D}_p^*(a, b) \cdot H(W)}{24} \leq h_F(E) \leq \frac{p}{6} \ln(abc).$$

Proof. By Faltings' arithmetic Noether formula, the stable Faltings height $h_F(E)$ of a semi-stable Frey scheme scales with the self-intersection number of the metric canonical line bundle $\overline{\omega}_{\mathcal{X}_0(N)/\mathbb{Z}}$ (Faltings, 1983; Silverman, 1986):

$$h_F(E) = \frac{1}{12} \widehat{c}_1 \left(\overline{\omega}_{\mathcal{X}_0(N)/\mathbb{Z}} \right)^2 + \mathcal{O}(\log(\mathrm{rad}(abc))) \, .$$

Integrating the volume form of the operator kernel across the discrete surface cancels out the modular curve's volume factor (Saito, 1988; Zhang, 1993). This isolates the lower floor. Concurrently, the geometric limits verified by Szpiro (1981) mandate that the Faltings height cannot exceed the minimal discriminant capacity (Hindry & Silverman, 2000; Vojta, 1987).

Combining the active floor with this definitive ceiling yields the stated bottleneck inequality. This relation remains strictly conditional on the explicit initialization matching coefficients of the canonical sheaf and is not presented as a direct, unconditioned consequence of absolute geometry. ☐

5 Evaluation of Boundary Obstructions and Modularity Tearing

We evaluate the complete framework across both coordinate boundaries to map out its definitive containment fields (Akhtar, 2026c, 2026d).

5.1 Localized Parity Obstructions

If the integer coordinates are perfectly balanced ($a = b$), primitive coprimality forces $a = b = 1$. This transforms the core equation into

$$1^p + 1^p = c^p \implies 2 = c^p,$$

which lacks integer solutions for any prime $p > 2$.

For coordinates where $a \neq b$ but share identical odd alignments ($a \equiv b \equiv 1 \pmod 2$), the minimal discriminant acquires an even valuation

$$v_2(\Delta_E) \geq 4p$$

(Frey, 1986). This structural property implies the local conductor is ramified at the prime $q = 2$. Ken Ribet's Level-Lowering Theorem states that the attached mod-p Galois representation $\rho_{E,p}$ can undergo a modular level-drop down to the target conductor $N' = 2$ if the representation is flat or unramified at 2 (Ribet, 1990; Serre, 1987).

Crucially, because exact parity alignments do not automatically cover or eliminate a full "near-balanced" real neighborhood where

$$a \not\equiv b \pmod 2,$$

this block is strictly localized. It handles the discrete isotropic baseline case and provides key modular insights but does not independently establish a broad, regional algebraic obstruction across non-aligned coordinate spaces.

5.2 The Skewed Boundary Obstruction ($w_a \to 0$)

Suppose a counterexample avoids the Galois block by being highly skewed ($a = 1$), implying

$$w_a = \frac{1}{1 + b^2}.$$

We evaluate the scale-insulated bottleneck inequality by dividing both sides by the shared integer volume factor $p \cdot \ln(abc)$ (Akhtar, 2026d):

$$\frac{\mathcal{D}_p(a, b) \cdot H(W)}{24} \leq \frac{1}{6}.$$

We take the global asymptotic limit as the prime exponent $p \to \infty$. The discrete closure defect functional evaluates smoothly to 1:

$$\lim_{p\to\infty} \mathcal{D}_p(a, b) = \lim_{p\to\infty} \left[1 - w_a^{p/2} - w_b^{p/2}\right] = 1 - 0 - 0 = 1.$$

Substituting this limit back into the bottleneck simplifies the expression to

$$\frac{H(W)}{24} \leq \frac{1}{6} \implies H(W) \leq 4.$$

We track the behavior of the remaining coordinate entropy $H(W)$ near the skewed boundary ($w_a \to 0^+$) (Cover & Thomas, 2006):

$$\lim_{w_a \to 0^+} H(W) = \lim_{w_a \to 0^+} [-w_a \ln w_a - w_b \ln w_b] = 0 - 0 = 0.$$

Substituting this boundary limit into the inequality yields

$$0 \leq 4.$$

Because $0 \leq 4$ is a completely true mathematical statement, the scale-weighted height bottleneck fails to force a contradiction. The geometric height floor shrinks faster than the allowed modular volume, proving the skewed coordinate region is stable and does not crash under pure analytic variation.

6 Conclusion

The comprehensive verification of the regularized APG framework demonstrates that continuous real analysis, when constrained to discrete modular channels, stabilizes perfectly at the boundaries. The scale-weighted functional successfully eliminates scale-blind spots but confirms that the analytic height floor collapses safely below the Szpiro ceiling under extreme coordinate skews.

Because the analytic flow engine remains stable in the skewed domain ($0 \leq 4$), it does not provide an independent proof of Fermat's Last Theorem. This containment boundary represents the core novelty and utility of Volume V: it locates the exact point of stabilization for continuous methods, reinforcing the conclusion that the rigid, discrete algebraic geometry of the original Ribet–Wiles framework remains the established pathway to eliminate semi-stable Frey schemes.

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